

Cat[®] Energy Control System (ECS)

Serial-Ethernet Modbus Gateway Configuration Guide

Cat[®] Part # 657-6662 (Serial to Ethernet Modbus Gateway)

Cat[®] ECS 100

Cat[®] ECS 200

Cat[®] ECS 300

Cat[®] ECS 400

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Table of Contents

1 DOCUMENT SCOPE..... 4

2 CONFIGURATION RESOURCES..... 4

3 CONFIGURATION STEPS..... 5

4 EXPORT/IMPORT 14

Serial-Ethernet Modbus Gateway Configuration Guide

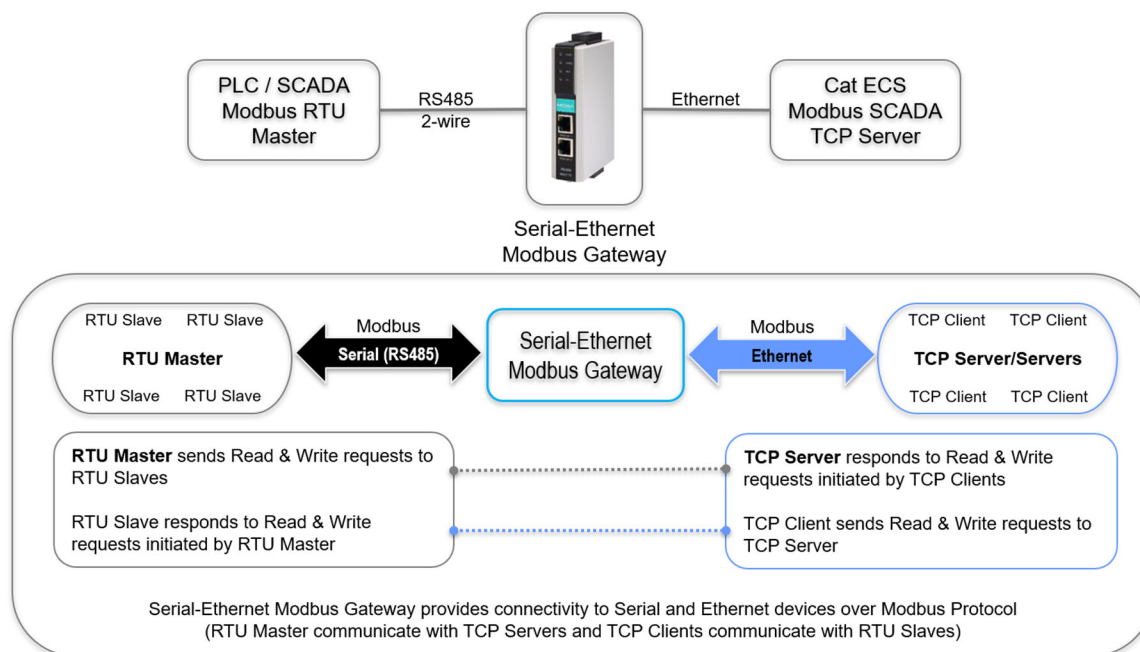
1 Document Scope

This document provides the standard configuration settings for MB3170I-T serial-to-Ethernet Modbus gateway to communicate with Cat® Energy Controls System (ECS) family of genset controllers via SCADA Modbus (see Cat part no 657-6662 specifications in [LEXE22903](#)).

The configuration is intended to use with applications where Modbus RTU Master PLC or SCADA systems communicating with Cat ECS genset controls over Modbus TCP Ethernet.

Note: This document is not intended to server as guide for Modbus implementation or RS485 wiring.

Serial-Ethernet Modbus Gateway



2 Configuration Resources

The MB3170I-T configuration requires the MGate™ Manager utility which can be downloaded from the original equipment manufacturer (OEM) product page. Go to Resources section of the product page. Under the Resources section, go to “Software, Firmware, and Drivers” and then click Utility. Download and install the latest version of MGate™ Manager Utility.

The Serial-Ethernet Modbus Gateway product manual can be downloaded from the product page. Under Resources section, go to “Support Documents” and click Manuals. Download the latest manual under this section.

Consult Serial-Ethernet Modbus Gateway manual for wiring, termination and other advanced configuration settings including Priority Control and System configurations.

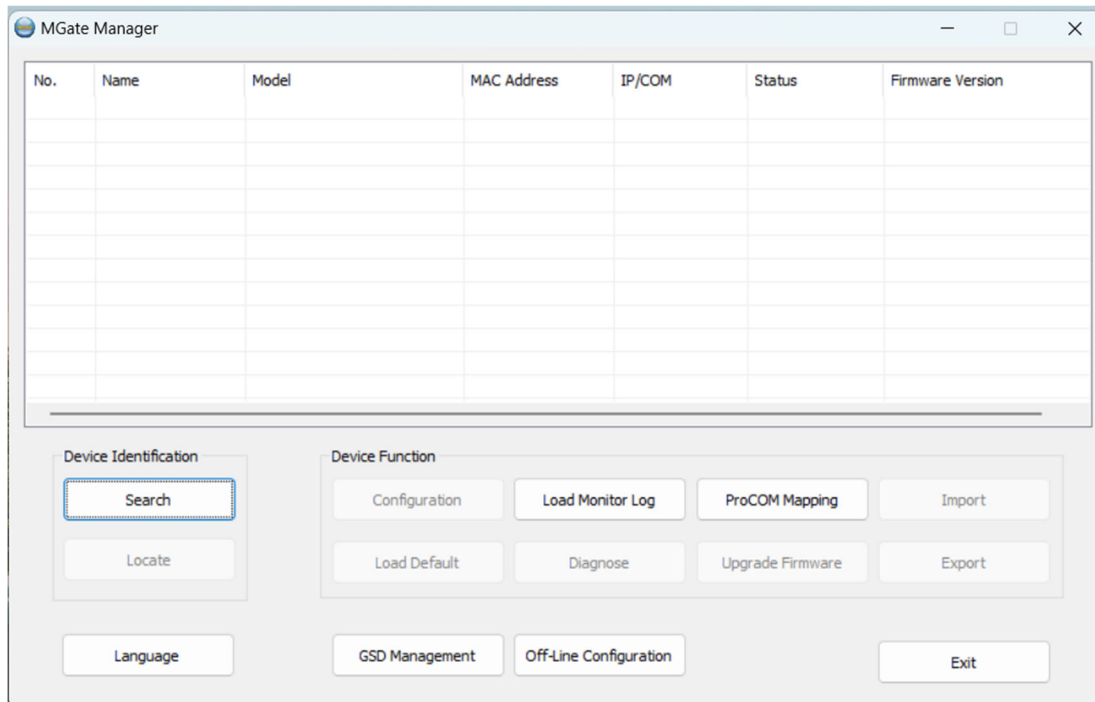
Note: The letter “I” in the model’s name “MB3170I-T” refers to Serial port with 2 kV isolation protection and letter “T” refers to -40 to 75°C wide operating temperature. The configuration steps apply to all variances of MB3170/MB3270 serial-Ethernet Modbus Gateways.

Serial-Ethernet Modbus Gateway Configuration Guide

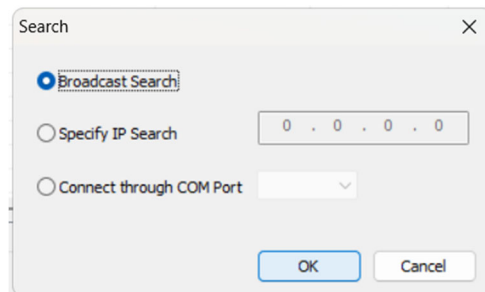
3 Configuration Steps

Connect your PC/Laptop Ethernet directly to one of the RJ45 port on MB3170I-T using an Ethernet cable.

Run MGate™ Manager Utility.

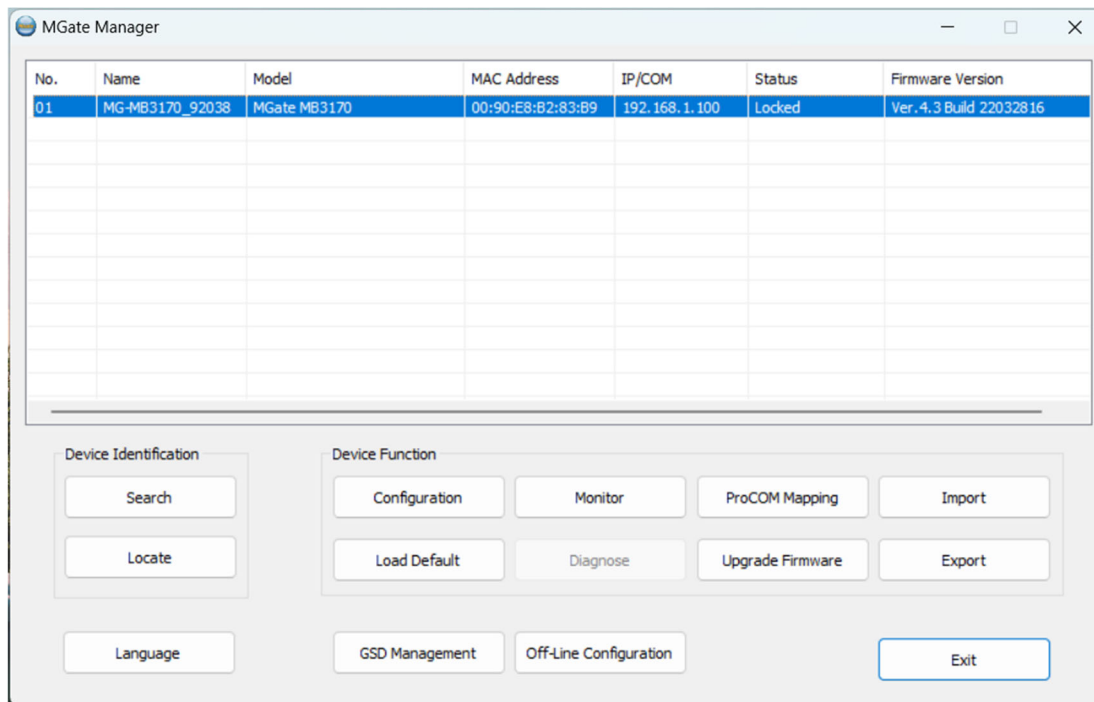


Click Search button, on popup window leave default selection “Broadcast Search” selected, click “Ok”.

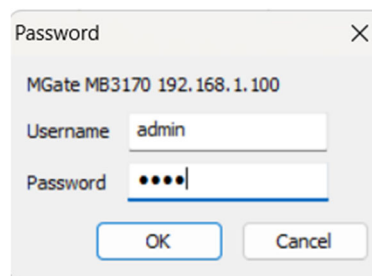


Serial-Ethernet Modbus Gateway Configuration Guide

MGate™ will search and display the device in the search window.

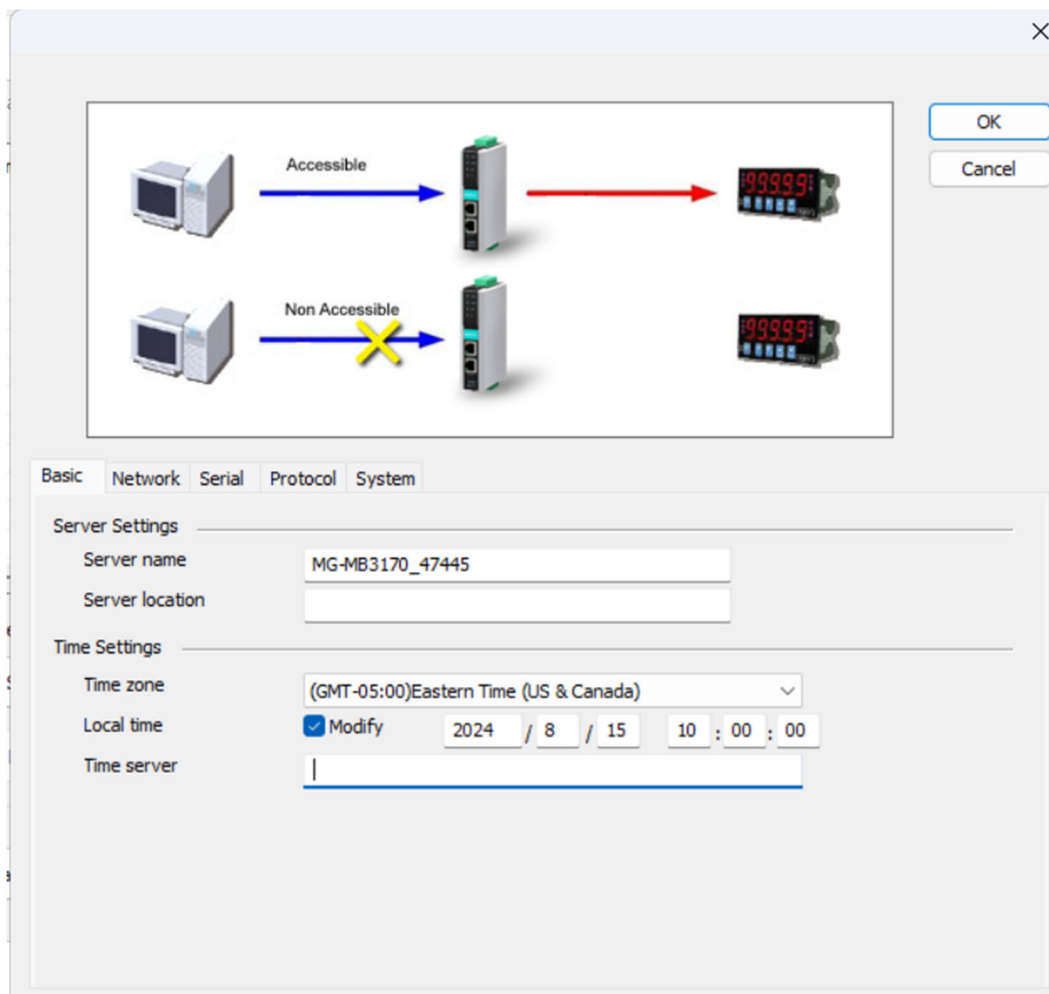


Select the device and double click it. It will prompt for the username and password. Use default username: **admin** and password: **moxa**



Serial-Ethernet Modbus Gateway Configuration Guide

Under Basic tab, configure Time Settings according to required time zone.



The image shows a configuration window for a Serial-Ethernet Modbus Gateway. The window has a title bar with a close button (X). Inside, there is a diagram illustrating the gateway's function. The diagram shows two scenarios: 'Accessible' and 'Non Accessible'. In the 'Accessible' scenario, a computer icon is connected to a gateway icon by a blue arrow, and the gateway is connected to a digital display icon by a red arrow. In the 'Non Accessible' scenario, a computer icon is connected to a gateway icon by a blue arrow with a yellow 'X' over it, and the gateway is connected to a digital display icon by a red arrow. To the right of the diagram are 'OK' and 'Cancel' buttons. Below the diagram is a tabbed interface with tabs for 'Basic', 'Network', 'Serial', 'Protocol', and 'System'. The 'Basic' tab is selected. Under the 'Basic' tab, there are two sections: 'Server Settings' and 'Time Settings'. The 'Server Settings' section has fields for 'Server name' (MG-MB3170_47445) and 'Server location'. The 'Time Settings' section has a 'Time zone' dropdown menu (set to '(GMT-05:00)Eastern Time (US & Canada)'), a 'Local time' section with a checked 'Modify' checkbox and a date/time picker (2024 / 8 / 15 10 : 00 : 00), and a 'Time server' text field.

OK
Cancel

Basic Network Serial Protocol System

Server Settings

Server name MG-MB3170_47445

Server location

Time Settings

Time zone (GMT-05:00)Eastern Time (US & Canada)

Local time ☒ Modify 2024 / 8 / 15 10 : 00 : 00

Time server

Serial-Ethernet Modbus Gateway Configuration Guide

Under Network tab, configure MB3170I-T device IP address associated with this MB3170I-T device according to your site-specific choice.

The screenshot shows a configuration window for the MB3170I-T device. At the top, there is a diagram illustrating the device's connectivity. The diagram shows an 'Ethernet' network (represented by a blue line) connected to a computer and a server. The 'Serial' network (represented by a grey line) is connected to four industrial devices. The MB3170I-T device is shown as a central unit with a padlock icon, indicating it is locked or secured. Below the diagram, there are tabs for 'Basic', 'Network', 'Serial', 'Protocol', and 'System'. The 'Network' tab is selected. Under the 'Network' tab, the 'Network Configure' dropdown is set to 'Static'. The 'IP Address' field is set to '192 . 168 . 1 . 100'. The 'Netmask' field is set to '255 . 255 . 255 . 0'. The 'Gateway' field is set to '0 . 0 . 0 . 0'. The 'DNS1' field is set to '0 . 0 . 0 . 0'. The 'DNS2' field is set to '0 . 0 . 0 . 0'. There are 'OK' and 'Cancel' buttons in the top right corner of the window.

OK
Cancel

Basic Network Serial Protocol System

Network Configure Static

IP Address 192 . 168 . 1 . 100

Netmask 255 . 255 . 255 . 0

Gateway 0 . 0 . 0 . 0

DNS1 0 . 0 . 0 . 0

DNS2 0 . 0 . 0 . 0

Serial-Ethernet Modbus Gateway Configuration Guide

Under Serial tab, configure serial side of the MB3170I-T device for RS485. This is where Modbus RTU Master will be connected

The screenshot displays the configuration interface for the Serial-Ethernet Modbus Gateway. The top section shows a diagram of the device connected to an Ethernet network and a Serial network. The Serial network is highlighted in red, indicating the current configuration focus. The Serial network is connected to a Modbus RTU Master (represented by a computer icon) and a Modbus RTU Slave (represented by a server icon).

The Serial tab is selected, showing the configuration for Port 1. The configuration parameters are as follows:

Parameter	Value
Port 1	
Baudrate	19200
Flow Control	None
Parity	None
FIFO	Enable
Stop bit	1
Interface	RS485 2-w
Data bits	8

The bottom section shows the DB9 Male and Terminal Block pin configurations. The DB9 Male pin configuration is as follows:

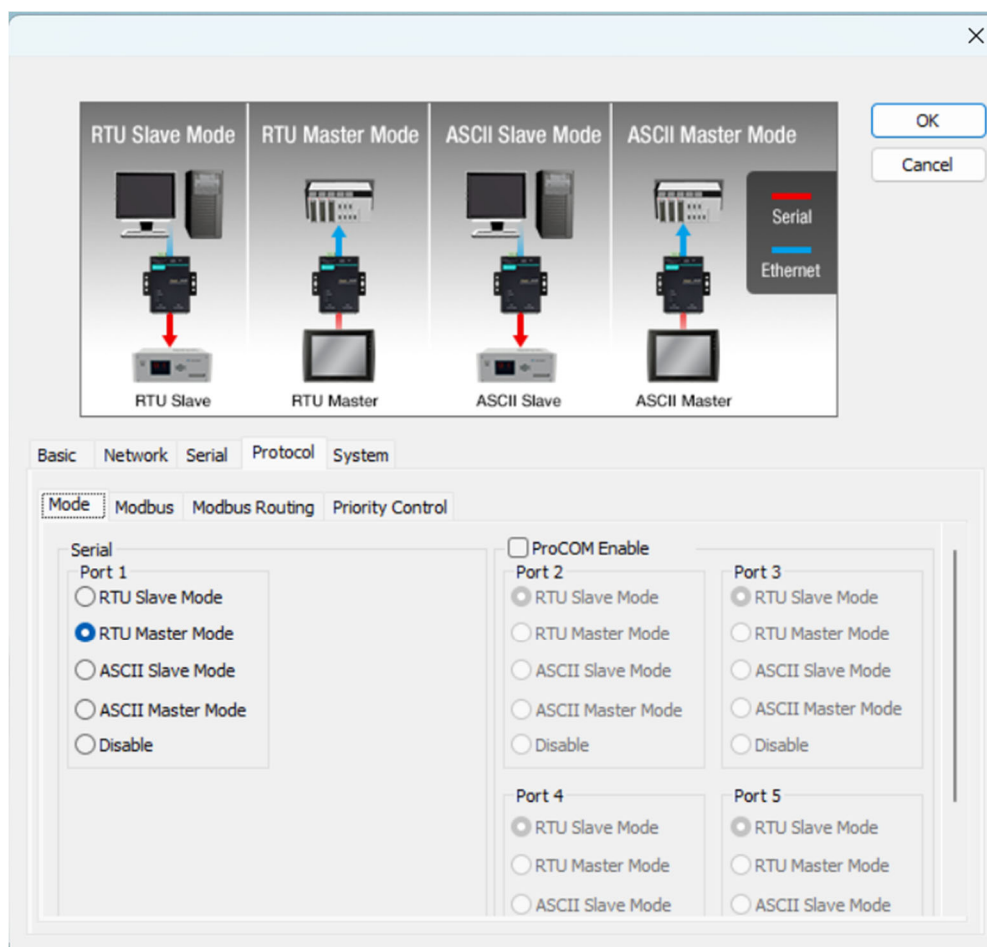
PIN	RS-232	PIN	RS-232
1	DCD	5	GND
2	RxD	6	DSR
3	TxD	7	RTS
4	DTR	8	CTS

The Terminal Block pin configuration is as follows:

PIN	RS-422/485 (4W)	RS-485 (2W)
1	TxD+	-
2	TxD-	-
3	RxD+	Data+
4	RxD-	Data-
5	GND	GND

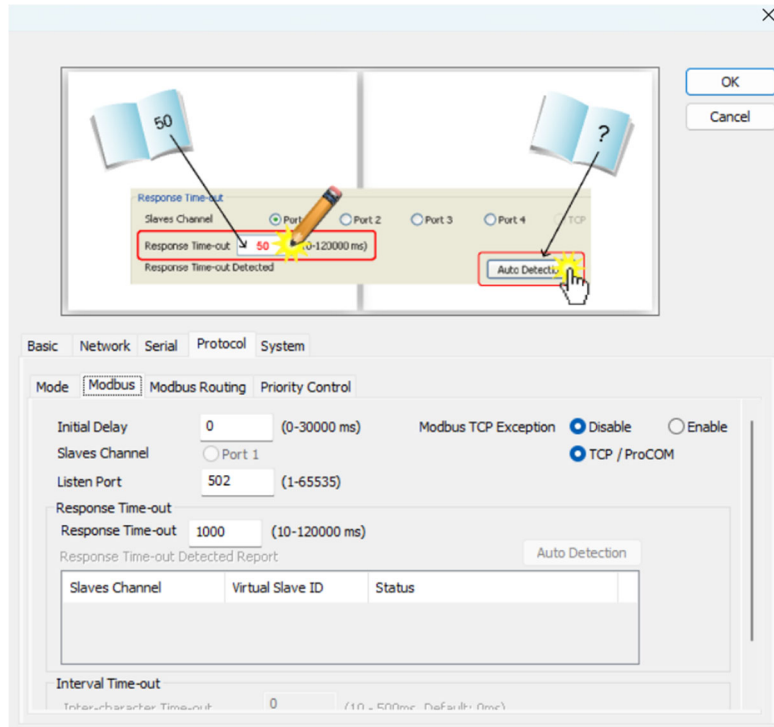
Serial-Ethernet Modbus Gateway Configuration Guide

Under Protocol sub-tab Mode, configure serial port mode for the device will be connected to the serial side of MB3170I-T. Configure Port 1 to “RTU Master Mode”. Uncheck “ProCOM Enable”.



Serial-Ethernet Modbus Gateway Configuration Guide

Under Protocol sub-tab Modbus, leave the default settings. Consult MB3170I-T manual for advanced settings per your specific need other than the default settings under this tab.



Serial-Ethernet Modbus Gateway Configuration Guide

Under Protocol sub-tab Modbus Routing, configure “Slave ID Table” and click add button.

- Set Remote IP Address to the Cat ECS SCADA IP address. Note: Cat ECS SCADA IP address can be configured on Cat ECS controller using Cat-ET or HMI.
- Set TCP Port to 502.
- Set Slave ID Offset to 0.

Cat ECS will require three (3) routing entries for each Read (TCP port 502), Event (TCP port 50201) and Write (TCP port 50200) for Cat Energy Controls System (ECS) SCADA Modbus corresponding to the same Cat ECS SCADA IP address. All three Slave IDs can vary based on site-specific application and will be used by the RTU Master to access the Cat ECS SCADA Modbus maps.

In the following example Slave ID 1 is used/mapped for Reads, Slave ID 2 for Writes and Slave ID 3 for Events.

The image displays three sequential screenshots of the 'Slave ID Table' configuration window, illustrating the setup for three different Modbus operations: Read, Write, and Event.

Top Screenshot (Read Configuration):

- Destination: Remote IP Address
- Remote IP Address: 192 . 168 . 1 . 6
- TCP Port: 502
- Slave ID Start: 1
- Slave ID End: 1
- Slave ID Offset: 0

Middle Screenshot (Write Configuration):

- Destination: Remote IP Address
- Remote IP Address: 192 . 168 . 1 . 6
- TCP Port: 50200
- Slave ID Start: 2
- Slave ID End: 2
- Slave ID Offset: 0

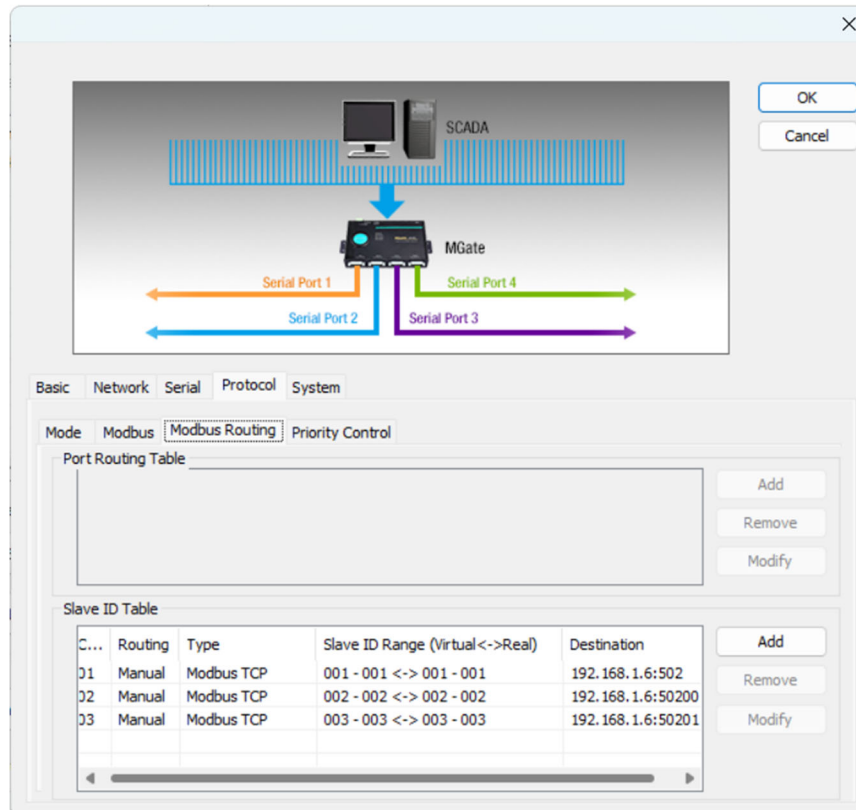
Bottom Screenshot (Event Configuration):

- Destination: Remote IP Address
- Remote IP Address: 192 . 168 . 1 . 6
- TCP Port: 50201
- Slave ID Start: 3
- Slave ID End: 3
- Slave ID Offset: 3

Serial-Ethernet Modbus Gateway Configuration Guide

Once finished, Slave ID Table will look like as shown below.

Note: Additional Slave ID routing can be added to the Slave ID Table for additional Genset Cat ECS controllers.



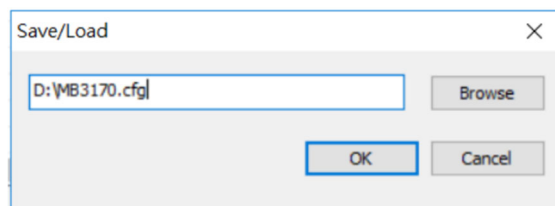
This completes the standard configuration of MB3170I-T Modbus Serial-Ethernet gateway.

4 Export/Import

The export function saves all the configuration settings and parameters of the MB3170 Modbus gateway in a *.cfg file. There are three main reasons for using the Import and Export functions.

- **Applying the same configuration to multiple units**
The Import/Export configuration function is a convenient way to apply the same settings to the devices. You can export the configuration as a “*.cfg” file and then import the configuration file onto other devices at any time.
- **Backing up configurations for system recovery**
The export function allows you to export configuration files that can be imported onto other gateways to restore malfunctioning systems within minutes.
- **Troubleshooting**
Exported configuration files can help your engineering support to identify system problems that provide useful information for Technical Service Team when maintenance visits are requested.

The configuration file can be exported/saved by selecting the device and then clicking the Export button. Enter an appropriate file name and use the Browse button to save the file to a specific path. Then, click the OK button. A confirmation message will pop up to say the configuration file is saved successfully as a “*.cfg” file.



Similarly, the saved configuration file can be imported into your target device by first selecting the target device and then clicking the Import button.

Wait for the MGate™ Manager to finish configuring the target device. If the import configuration is done successfully, a confirmation message will pop up. After closing the message dialog, the MGate™ Manager will automatically execute a Broadcast Search for all MGate™ units on the LAN. Your MGate™ device should reappear in the list of units.

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